ENGINEERING TECHNICS



ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan

Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ENGINEERING TECHNICS

GE PROTECTION RELAYS

Multilin 345
The Multilin™ 345 is a member of the Multilin 3 Series protective relay platform and has been designed for the protection, control and management of power transformers as primary or backup protection device
Comprehensive metering
Event Recorder: 256 events (1ms time stamping)
Programmable oscillography up to 32 samples per cycle and digital states and Fault Report
4X20 character LCD display
Control panel with 12 LED indicators
Relay health diagnostics and Breaker monitoring
Security and password control
SNTP or IRIG-B time synchronization







Protection Relays

Line Differential and Distance Protection
MiCOM Agile P54x Models
- *P543*
- *P544*
- *P545*
- *P546*

MiCOM Agile P640 Series
Transformer Protection Relays
- *P642 2 Winding transformer management IED*
- *P643 3-End transformer management IED*
- *P645 5-End transformer management IED*

MiCOM Agile P74x Series
Distributed Busbar Differential Relays
- *P741 Central Unit ,*
- *P742 Peripheral Unit ,*
- *P743 Extended Peripheral Unit*

MiCOM Agile P746
Centralized Busbar Differential Relay
Distance Protection Relays
P441, P442, P443, P444, P445, P446

P40 Agile
Feeder Management

The P14N non-directional feeder, P14D directional feeder and P94V voltage and frequency IEDs bring the design rigor and technology of transmission applications to the entire utility and industrial market.

P14NB	*P14N*	*Non-directional feeder*	*20TE*
P14NL	*P14N*	*Non-directional feeder with auto reclose*	*20TE*
P14DB	*P14D*	*Directional feeder*	*20TE*
P14DL	*P14D*	*Advanced directional feeder with auto reclose and check synchronizing*	*20TE*
P94VB	*P94V*	*Voltage and frequency*	*20TE*
P94VP	*P94V*	*Voltage and frequency with auto re close and check synchronizing*	*20TE*





ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan
Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ELPROM HEAVY INDUSTRIES

ELPROM MAIN PRODUCTS

POWER AND DISTRIBUTION TRANSFORMERS from 2.5 up to 300 MVA and up to 400 kV

ON-LOAD TAP-CHANGERS up to 1600 A, 300 kV

OFF-CIRCUIT TAPCHANGERS up to 2000 A, 300 kV





The EHI OLTCs meet the requirements of the IEC 60214-1: 2003 standard.

Basic technical data

OLTC type	***RS 9.3 III 200***	***RS 9.3 III 400***	***RS 9.3 III 630***	***RS 9.3 I 200***	***RS 9.3 I 400***	***RS 9.3 I 630***	***RS 9.3 I 800***	***RS 9.3 I 1200***	***RS 9.3 I 1600***
Number of phases and application		*3 (in the neutral)*				*1 phase*			
Maximum rated through current (A)	*200*	*400*	*630*	*200*	*400*	*630*	*800*	*1200*	*1600*
Short circuit withstand current (κA)									
r.m.s. value, 3sec.duration	*4*	*8*	*12.6*	*4*	*8*	*12.6*	*16*	*24*	*24*
peak value	*10*	*20*	*31.5*	*10*	*20*	*31.5*	*40*	*60*	*60*
Maximum rated step voltage per phase (V)	*4000*	*3500*	*3000*	*4000*	*3500*	*3000*	*3000*	*2500*	*2000*
Rated switching capacity (kVA)	*800*	*1400*	*1890*	*800*	*1400*	*1890*	*2000*	*3000*	*3200*
Rated frequency (Hz)	*50 /60*	*50 /60*	*50 /60*	*50 /60*	*50 /60*	*50 /60*	*50 /60*	*50 /60*	*50 /60*

OLTCs type RS9 are with elliptical upper flange. OLTCs type RS9.3 are with round upper flange. All the rest technical data are the same.

ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan

Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

HIKMICRO

THERMOLOGY PRODUCTS / THERMOVISION CAMERAS



G60 / G40

Key Features

Thermal resolution: 640 x 512 (327 680 pixels)
NETD : < 35 mK (@ 25 °C,F#=1.0)
Temperature measurement range : -20 °C to 650 °C (-4 °F to 1202 °F)
Accuracy: Max (±2°C/3.6°F, ±2%)
Measurement Presets: Center spot, Hot spot, Cold spot, User Preset Point/Line/Area
Auto/Manual/Laser focus
50 Hz image frequency
4.3" LCD touch screen display
8MP Optical camera
Up to 4 hours continuous running

Storage

Storage Media Removable Micro SD Card (64 GB)
Image Storage Capacity 60 000 Images
Image File Format BMP with Measurement Data Included
Video Storage Capacity 54 Hours

M30

Specifications:

Field of view FOV 37.5° × 28.5°
Digital Zoom 1x / 2x / 4x / 8x
Picture Mode Thermal, Visual, Fusion, Picture-in-Picture
Storage media Removable Micro SD Card 16GB
Built-in Laser Pointer Class II
Wi-Fi 802.11 a/ac/b/g/n (2.4 and 5 GHz)
Bluetooth 4.2
Voice comment up to 60 seconds
Battery life 5 hours
IP Class IP54
Interface USB TYPE-C
Weight ± 660 grams
Dimensions 244mm × 100mm × 104mm

Key features:

Thermal resolution 384 x 288 (110,592 Pixels)
Thermal sensitivity of <0.04°C (NETD <40mK)
Manual Focusable lens (minimum focus 10cm)
Temperature measurement range 20°C - 550°C
Absolute accuracy ±2°C or ±2%
8MP Visual camera with Bi-Spectrum Blending
25Hz fast frame rate
3.5' LCD touch screen
Pre-sets for user-defined measure-ments
Center spot, hotspot, cold spot,



Mini Series

Thermal Module

Image Sensor Vanadium Oxide Uncooled Focal Plane Arrays
IR Resolution 160 × 120 (19 200 pixels)
Spectrum Range 8 - 14 µm
NETD < 40 mK (@ 25 °C, F#=1.0)
Focal Length 3.2 mm
FOV (Field of View) 50° × 38° (H × V)
Minimum Focus Distance 20 cm
Focus Fixed
Image Frequency 25 Hz
Aperture F1.1

Thermography Function

Accuracy
Max (±2°C/3.6°F, ±2%), applicable 60 sec after start-up when the ambient temperature 15°C to 35°C (59°F to 95°F) and object temperature above 0°C (32°F)
Object Temperature Range -20 °C to 350 °C(-4°F to 662°F)



ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan
Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ELSTER ELECTRICITY



A1800 ALPHA meter is a highly accurate, robust meter suitable for advanced commercial, industrial and substation metering applications

TECHNICAL DETAILS

- *IEC 62053 accuracy Class 0.2 %, 0.5 %, and 1.0 %*
- *UV-protected, polycarbonate, IP54 enclosure*
- *Wide operating ranges: 46 V to 528 V, 1 mA to 10 A (CT connected)*
- *Temperature: -40 °C to +85 °C*
- *Energy plus demand for kWh, kVARh and kVAh*
- *Four-quadrant metering for export import*
- *Up to 4 tariffs per day for 4 day types, multiple seasons*
- *Nonvolatile memory*
- *Easily replaceable battery located under the terminal cover*
- *Precision internal clock with backup timekeeping provided by super capacitor and long-life battery*
- *16-segment character LCD optional backlight*
- *Windows-based software with multi language support*
- *Optional Read without Power feature ensures the meter can be read when it is not externally powered*



Elster AS1440 The alpha meter is available either for direct or CT/VT connection. The meter is in accordance with the relevant DIN, MID and IEC standards.

- *High accuracy and stability*
- *4-Quadrant measurement*
- *8 energy and 4 demand tariffs, independently controllable*
- *measurement of active, reactive and apparent demand*
- *integrated tariff clock*
- *time back-up with super capacitor + internal battery*
- *readout of meter data without main power*
- *Optical interface acc. EN 62056-21*
- *Electrical interface (CL0, RS232, RS485)*
- *measuring of instantaneous values*
- *profile of instrumentation values*
- *user friendly reading, setting and programming tool alphaSET*



Elster A1100 Electronic Polyphase Meter

Features

- *Accuracy Class 1 or Class 2*
- *kWh import or kWh import/export*
- *3 phase, 4 wire or 3 phase, 3 wire*
- *Large figure display*
- *Compact design*
- *Double insulated, glass filled polycarbonate case to DIN 43857 Part 2 and Part 4*
- *IP53 in accordance with IEC 60529:1989*



Elster A100C Electronic Single Phase Meter

Features

- *Accuracy Class 1 or Class 2*
- *kWh import or kWh import/export*
- *Large digit (9.8mm) multilingual display with chevron information indication*
- *Compact design (130mm Wide x 97mm High x 47mm Deep)*
- *Double insulated, glass filled polycarbonate case*
- *Permanently fixed main cover*
- *IP53 in accordance with IEC 60529:1989*

ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan

Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ENGINEERING TECHNICS

Power FULL STOP
... nothing else

DRY TYPE CAST RESIN TRANSFORMERS

Cast resin transformers represent an alternative solution for electric power distribution in medium voltage applications.



*The standard product range from **POWER. Sp. z o.o.** covers distribution cast resin transformers from 50kVA up to 5000kVA, with insulation class up to 36kV, with Eco-design, reduced or standard losses.*

All of our transformers are manufactured in compliance with IEC 60076-11 standard and in particular meet the requirements for climatic (C), environmental (E) and fire (F) classes.

Each and every transformer undergo a series of routine tests in compliance with IE-C60076-11 at our own testing room

Minimal fire risk

The materials used for production of cast resin transformers are hardly inflammable and can be defined as self-extinguishing, which is why there is no need to use any special fire protection coatings for this type of transformers. Furthermore, in case of fire, the emission of toxic gases and opaque fumes is very low. These advantages make cast resin transformers a perfect choice for installations at places where safety is crucial, such as hospitals, public premises, airports, subway lines, mines, oil rigs, nuclear power plants, ships, etc.

No cooling liquids

Cast resin transformers need only air for cooling. They are therefore nearly maintenance-free, whereas a liquid-cooled transformer needs much more attention. This also makes cast resin transformers eco-friendlier since no liquid coolants – of whichever chemical composition – can be released into the environment.





Advantages of IP enclosure

Instead of traditional transformer bays, simple IP enclosures can be used for access prevention and protection of cast resin transformers. IP enclosures come in different types and colours depending on the client's needs and preferences.

High short-time overload capability

Current density in the windings of cast resin transformers is considerably lower than in those of oil-immersed transformers. Hence, short-time load peaks, such as with wind power installations, can be easily overcome with no need to plan adequate oversizing.



ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan

Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ENGINEERING TECHNICS

MIDEL® SAFETY INSIDE

SYNTHETIC ESTER TRANSFORMER FLUID



MIDEL eN 1204
Natural Ester Transformer Fluid (Rapeseed) manufactured using renewable vegetable oil, readily biodegradable, providing fire safety and environmental protection from a natural ester.

MIDEL eN 1215
Natural Ester Transformer Fluid (Soya) manufactured using renewable vegetable oil. Readily biodegradable, providing fire safety and environmental protection from a natural ester.

MIDEL 7131
Synthetic Ester Transformer Fluid Fire safety and improved performance from the original ester transformer fluid

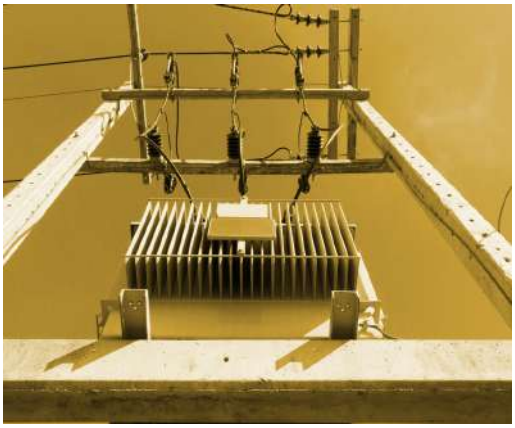
MIDEL offers more than the benefits of a fire-safe fluid – there are also clear advantages to be gained in cost and risk reductions, deployment innovations, corporate social responsibility, and asset performance.

MIDEL 7131 is the only synthetic ester approved by both FM Global and UL.





MIDEL is the acknowledged leading brand of ester transformer fluid. Since the 1970s MIDEL has been used in hundreds of thousands of installations worldwide, providing unrivalled fire safety, environmental protection, increased asset performance and real cost savings.



ENGINEERING TECHNICS
67-A, New Muslim Town, Lahore-Pakistan
Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ACCESSORIES FOR POWER TRANSFORMERS



Temperature Indicator
Standard: IEC 60076-22-1 and JB/T 6302-2005
Ambient temperature From -40°C up to +70°C
Degree of protection IP66
Ventilation valve 2; to prevent the formation of condensation
Cable gland Up to 3; M25x1.5
Micro switches Up to 4
Supply voltage 24 V dc ±10% polarized
Power consumption 0,5 W
Measuring range eWTI: 0°C/+150°C; 0°C/+160°C; 0°C/+180°C; -50°C/+150°C
eOTI: -50°C/+150°C; -20°C/+130°C; -20°C/+140°C; 0°C/+120°C; 0°C/+150°C; 0°C/+160°C
Measuring tolerance 1.5% of full scale
Accuracy switch 1.5% of full scale

Porcelain Bushings

Application

Oil-filled power transformers
Suitable for new installations and retrofits

Technical data at a glance

Nominal voltage: 1 – 52 kV
Rated current: 250 – 8000 A
Compatible with EN and DIN standards







Pressure Relief Device

Standard IEC 60076-22-1
Ambient temperature -40°C up to +80°C
Switches Up to 3, Changeover
Cable gland M25x1.5 – nickel plate brass
Degree of protection IP65
Power supply 24 V dc ±10% polarized Analog output 4-20 mA

Buchholz Relays

Standard IEC 60076-22-1 IEC 60076-22-1 IEC 60076-22-1
Oil temperature From -40°C up to +120°C
Ambient temperature From -40°C up to +80°C
Max. nominal current 2A
Min. switching current 10 mA/24 V dc
Flow vents speed 10 m/s to 3.0 m/s (each +- 15%)
Cable gland 1; M25x1.5
Degree of protection IP65

Breathers

Standard IEC60076-22-7
Ambient temperature From -40 °C to 80 °C (from -40 °F to 176 °F)
Pressure load loss of air through the dehydrating breather 0,003 kg/cm2 for in-let air; 0,005 kg/cm2 for out-let air



ENGINEERING TECHNICS
67-A, New Muslim Town, Lahore-Pakistan
Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ENGINEERING TECHNICS

Axial Transformer Cooling Fans



Axial Transformer Cooling Fans

Engineered for transformer operation
Fully closed motor up to ip66
Easy installation
High corrosion resistance
Safe operation with complete guard grilles
Suitable for harsh ambient temperatures
Wide customization options
Eco design compatible

Standard IEC 60034-1, IEC 60335-2:80, ISO 5801
Balance Quality G6.3 according to ISO 1940-1
Tolerance Class 2 according to DIN 24166
Mounting Position Horizontal & Vertical
Motor Asynchronous motor with aluminum body
Winding Insulation Class F or H
Winding Temperature Rise Class B
Operation Indoor & Outdoor
Corrosion Class C3 / C4 / C5 - Medium / High acc. to ISO12944-2
Fasteners Stainless Steel





Model	A40C –F		A40C-B		A40C-S		A63C-F		A80C-B	
Type	A40C –F5	A40C –F6	A40C-B5	A40C-B6	A40C-S5	A40C-S6	A63C-F5	A63C-F6	A80C-B5	A80C-B6
Frequency	50Hz	60Hz	50Hz	60Hz	50Hz	60Hz	50Hz	60Hz	50Hz	60Hz
Supply Voltage	220-240V Δ / 380-420V Y	240-280V Δ / 400-480V Y	220-240V Δ / 380-420V Y	240-280V Δ / 400-480V Y	220-240V Δ / 380-420V Y	240-280V Δ / 400-480V Y	220-240V Δ / 380-420V Y	240-280V Δ / 400-480V Y	220-240V Δ / 380-420V Y	240-280V Δ / 400-480V Y
Nominal Current	1,2 A / 0,7 A	1,0 A / 0,6 A	0,9 A / 0,5 A	0,9 A / 0,5 A	0,8 A / 0,5 A	0,9 A / 0,4 A	3,2 A / 1,8 A	2,9 A / 1,7 A	3,1 A / 1,8 A	2,9 A / 1,7 A
Input Power	0,29 kW	0,29 kW	0,13 kW	0,19 kW	0,10 kW	0,13 kW	0,86 kW	0,89 kW	0,68 kW	0,73 kW
Speed	1440 rpm	1728 rpm	960 rpm	1152rpm	720rpm	864rpm	1440 rpm	1728 rpm	720 rpm	864 rpm
Sound pressure	73 dB(A) / 68 dB(A)	78 dB(A) / 73 dB(A)	63 dB(A) / 58 dB(A)	68 dB(A) / 63 dB(A)	56 dB(A) / 51 dB(A)	61 dB(A) / 56 dB(A)	79 dB(A) / 74 dB(A)	84 dB(A) / 79 dB(A)	71 dB(A) / 66 dB(A)	76 dB(A) / 71 dB(A)
Airflow	1,20 m³/s	1,20 m³/s	0,81 m³/s	0,93 m³/s	0,54 m³/s	0,71 m³/s	3,40 m³/s	3,42 m³/s	4,33 m³/s	4,43 m³/s
Weight	16 kg		16kg		17kg		31kg		49kg	

ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan

Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

SW INDUSTRY CO.



Transformer Radiator

Header Pipe Type (Welded Header)
Width: 300mm(12"), 310mm, 380mm (15"), 450mm(18"), 480(18.9"), 520mm (20.5"), 535mm (21")
Material (Thick.) : CR Steel, Stainless Steel (1mm, 1.2mm)
Welding: O2, CO2
Fin Distance (Pitch): 40, 45, 50mm
Surface Treatment:: Plain (Bare Metal), Customizing Painting, Hot Dip Galvanizing

Header Panel Type (Integral Header)
Width: 225mm(9"), 300mm(12"), 450mm(17.7")
Material (Thick.): CR Steel, Stainless Steel (1mm, 1.2mm)
Welding: O2, CO2
Fin Distance (Pitch): 40~50
Surface Treatment: Plain (Bare Metal), Customizing Painting, Hot Dip Galvanizing

Swan Neck Type (Goose Neck Type / Sky Type)
Width: 300mm(12"), 310mm, 380mm(15"), 450mm(18"), 480(18.9"), 520mm(20.5"), 535mm (21")
Material (Thick.): CR Steel, Stainless Steel (1mm, 1.2mm)
Welding: O2, CO2
Fin Distance (Pitch): 40, 45, 50mm
Surface Treatment: Plain (Bare Metal), Customizing Painting, Hot Dip Galvanizing

Corrugated Fins

Transformer Wheels

Radiator Valve
Butterfly Valve(Radiator Valve)

KP1206 / KP1207 / KP1210-1

Model	Type	A	B	ØC	ØD	ØE	ØF	G	ØH	I	J
KP1206	3"F	196	150	150	92	98	115	2.5	43	1.6.1	28
KP1207	4"F	218	172	180	110	122	146	2.5	45	127.3	28
KP1210-1	3"B	196	150	161.2	92	98	115	2.5	43	114	-

Usage: Pipe Connection area for oil immersed transformers
Installation: Stud Bolts type
Body FC 20
Option Item: Stud Bolts



Transformer Cooling Fan
(General Specifications of Transformer Cooling Fan)

Motor

Weatherproof enclosure (totally enclosed) a shaft water slinger and 4 x M5 drain holes to allow condensation to drain, regardless of the mounting position.
Class F insulation with Class B temperature rise for longer motor life.
Sealed ball bearings permanently lubricated with heavy duty grease rated -540°C to 1940°C
Motors are painted with 75 microns of high quality, gray (Munsell no. 5Y 7/1), outdoor epoxy.



Guards & Housings

The Fans feature basket type guards. They feature a high efficiency, venturi-type housing.
All fans are fully guarded and meet OSHA requirements.
Terminal Box connection Type
Painted, Hot dip galvanized or Stainless Steel guards are available on request.



ENGINEERING TECHNICS
67-A, New Muslim Town, Lahore-Pakistan
Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ENGINEERING TECHNICS

ACCESSORIES FOR TRANSFORMERS

ARES TRAFO



Ares Trafo produces the most suitable and fastest products and accessories for all transformer manufacturers that it works with in the production of transformer sector .

Transformer Bushings	*Breathers*
Transformer Valves	*Axial Cooling Fans*
Pressure Relief Valves	*Oil Lever Indicator*
Buchholz Relays	*Thermometers*



Bimetal Thermometers:

Temperature sensing from -10°C to 120°C
Adjustable limit switches (min / max)
Resettable indicator
Visible screen made of UV resistant polyamide
Protection class IP 65
Corrosion resistant materials

Magnetic Oil Level Indicator

Flat Gasket	*NBR LP 1100*
Arrow	*Brass Ms58*
Dial Plate	*Aluminum*
Front Glass	*Polycarbonate 143R*
Float Lever	*Brass Ms 70*
Main Body	*Aluminum*

Buchholz relay DN80 for Transformers EN 50216-2

ARES Buchholz Relay operates as a protection device that can be provided with two contacts for alarm and tripping.
Reed switches for detection of gas formation and dropping oil level
Flap valve to protect in case of excessive oil flow rate
Switch elements separated from the oil circuit
Mechanical testing of contact switches (test button)
Buchholz Relay MBP in accordance with EN 50216-2
Nominal pipe diameter: DN 80 (3")
Connection: Flange connection
Features: Reed switches up to 2 NO or 2 CO contacts per function , Ambient temperature: -30 °C bis 55 °C, IP65

Pressure Relief Device

Type: T – 10 short
Connections Diameter: R1" – R1 ½" – R2"
Working pressure range (bar): 0.20 – 0.70
for Distribution transformers
Body of valve: Cu Zn39 Pb3
Cap of valve: Cu Zn39 Pb3
Protection cap: Plastic- Pa6
Stud bolt: M 5x60 A2
Spring: Ø19x2x24
Washer: Cu Zn39 Pb3
Nut: M5-A2 DIN 934
O-Ring or Flat Gasket: NBR70





Bushings

DIN 42530 - EN 50386

ARES bushings 1kV bushing series according to DIN42530 and EN50386 with a rated current from 250A to 5000A, fully interchangeable with DIN standard for easy replacement. Can be Shipped pre-assembled or fully assembled, special designs can be provided for external environmental conditions, ensuring a longer service life and reducing leakage issues we can supply different gaskets such as NBR, VITON, FVMW -60 or CORK.

52kV bushings

NBR , Viton, Cork, FVMQ gaskets

52kV 1000A DIN 42533
52kV 2000A DIN 42533
52kV 3150A DIN 42533
metalized or unmetallized.

3kv 1000A Bushing DIN 42539

3kV 250A DIN42539
3kV 630A DIN42539
3kV 1000A DIN42539
3kV 2000A DIN42539
3kV 3150A DIN42539
3kV 4500A DIN42539



ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan

Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ENGINEERING TECHNICS

RANCAN

Hi-tech Laminated Wood for all industrial applications



Rancan Srl, established in 1960, is a private owned company, leader in the production of Laminated Densified Wood, which is used in all industrial sectors. In more than 50 years of history, the company has reached top rated standards of quality, boasting among its customers Groups and companies leaders in the Italian and International market.

RANPREX®, red beech laminated densified wood is made with the best wooden essences, impregnated with patented thermosetting resins and pressed at very high pressure and temperature. The result is a dense and very strong material, with different specific weights (densities), suitable for many different applications. It is also an excellent electric and thermal insulator, boasting good physical and mechanical characteristics of compression, flexion, and impact resistance. Moreover, it is self-lubricating, extremely resistant to wear, and it owns an excellent resistance to high and low temperatures.







ELECTROMECHANICAL INDUSTRY

RANPREX® is widely applied in the Electrical Oil-Immersed Power & Distribution Transformer Industry.

It complies with DIN7707 and IEC61061 norms and boasts:

- *Excellent dielectric properties*
- *High oil absorption values*
- *Excellent compressive strength*
- *Aptness to work at high temperature*
- *Consistent dimensional stability*
- *Different specific weight according to the application*

Standard Plates Sizes:

- *- 2500x1220*
- *- 2500x1100 (for container loads)*
- *- 2000x1200*
- *- 2200x1200*
- *- 2050x1100*

Thicknesses from 10 up to 300mm, and more.

	Tensile Strength			Standard Dimensions			
	Nut 1xth (N)	Nut 2xth (N)	Tensile Strength Threaded Rods (N)		L1 mm	L mm	Th mm
M8	2500	4300	2000	M8	14	20	10
M10	3400	6400	4000	M10	19	27	10
M12	6000	11500	6000	M12	24	34	12
M14	6200	17000	8000	M14	28	40	14
M16	11000	20500	11000	M16	32	45	16
M20	16000	27000	16000	M20	36	51	20







ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan

Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ENGINEERING TECHNICS

INNER MONGOLIA CHEN-FIR ELECTRICAL INSULATING MATERIALS

Be made up of 100% pulp vitriol conifer wood, the "OASIS" thin insulating paperboard has been titled "Good Quality" by the Light Industry Ministry and enter into the "Data of Famous product in China". It has excellent mechanical strength, high density, good compressed performance, strong anti-electrical ability etc.

- *Transformer Layer Insulating Paper*
- *High Voltage Cable Paper*
- *Pre-Compressed Pressed Board & Insulating Paper*
- *Diamond Dotted Paper*
- *Phenolic Paper Laminated Sheets & Rods*





Press paper

It be mainly used in the clapboards of transformers and other insulating components such as insulating tube. It can also be used as the board in electrical usage, such as high/low switch and the insulating material of home electrical equipment

Specification:

Thickness: from 0.1 to 0.6mm
color: cyan, red, blue ,natural
*package: sheet(1150mm*800mm) or roll (width:1000mm)*
Of course, any other specified customized ones can be made.

Press Board

It be used widely in the electrical insulating material machinery, electrical equipment, transformer

Specification：

Thickness: from 0.8 to 4.0mm
Color: natural, red,
*Package:sheet (1000mm*2000mm/400mm*3100mm) any other specified customized ones can be made.*



ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan

Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ELEGTRICAL INSULATION

Kraft Paper

Kraft Paper is made of 100% pure sulphate cellulose. It is used as insulation of high voltage parts in electrical equipment constructions. Ideal for wrapping wires, cables and as insulation in oil filled-transformers. Performs excellently on automatic wrapping machines.

ISOKRAFT Paper 3588
Standard thickness in mm 0.05 / 0.075 / 0.08 / 0.125
Standard width for all thicknesses in mm approx. 1,000 / 1,100

ISOKRAFT Paper 3688
Standard thickness in mm 0.05
Standard width for all thicknesses in mm approx. 1,000

ISOKRAFT Paper 3689
Standard thickness in mm 0.05 / 0.075
Standard width for all thicknesses in mm approx. 1,000 / 1,100

ISOKRAFT Paper 3693
Standard thickness in mm 0.05
Standard width for all thicknesses in mm approx. 1,000 / 1,100







Isovolta Paper

ISO DD Paper 3588
Standard thicknesses in mm 0.050 / 0.080 / 0.125
Standard width for all thicknesses in mm 960 / 1,100

ISO DD PAPER 0410
Thickness in mm 0.05 – 0.76
Width in mm 900

ISO DD PAPER E56
Thickness in mm 0.13 – 0.51
Width in mm 900

ISOCREPE PAPER 0410
Base Nomex® thickness in mm 0.05 / 0.08
Width in mm 900

SOCREPE PAPER 0411
Base Nomex® thickness in mm 0.18
Width in mm 900

ISO DD Paper 3040
Standard thicknesses in mm 0.130 / 0.200 / 0.250
Standard width for all thicknesses in mm 960 / 1,100



CALMICA® and CALMICAGLAS®
thermal class F-H (155 °c - 180 °c), thermosetting materials for main insulations, based on a mica paper fully impregnated with high temperature resistant epoxy resin.

CALMICA® 70 0900	*Pet film carrier with calcined mica, for coils and bars of high voltage machines*
CALMICA® 0867	*Pet film carrier with uncalcined mica paper, for coils and bars of high voltage machines*
CALMICA® S100 3052	*calcined mica paper on shrinkable Pet film carrier, for coils and bars of high voltage machines*
CALMICAGLAS® 0409	*standard type but with calcined mica. calcined mica, for highest output and voltage*
CALMICAGLAS® 0893	*With uncalcined mica, for coils and bars of low- and high-voltage machines*



Auxiliary Materials
ISOAD TAPE 2400 is based on a polyester film carrier reinforced with glass fabric. It is coated with a thermosetting acrylic adhesive which is suitable for applications in oil filled-transformers, e.g. as core-, layer- and final-insulation of coils.

ISOAD TAPE 4000 is based in a support of Nomex® Paper, and an acrylic thermosetting adhesive.

ISOTAPE GC is a Glass cloth flat tape made of twisted thread, without finish or primer coating .

ISOTAPE GP is a Woven glass/ polyester tape without finish. The polyester fibres are in the warp direction and glass fibres are in the weft direction

ENGINEERING TECHNICS
67-A, New Muslim Town, Lahore-Pakistan
Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ENGINEERING TECHNICS

ANHUI WONDERFUL-WALL COLOR COATING ALUMINIUM SCIENCE TECHNOLOGY CO., LTD.

ALUCOSUPER, founded in 1996, is the subsidiary of Jinggong Holding Group Co., Ltd, which's one of top 500 manufacturing enterprises in China. The company specializes in production of Color Coated Aluminum Coil/ Sheet, PCM, VCM, ACP, solid aluminum sheet etc.. After more than 20 years development, Alucosuper® has become one of the leading brands in the industry.



PCM PPGI Color Coated Galvanized Steel Coil



PPGI for Household Appliance Color Coated Steel Sheet



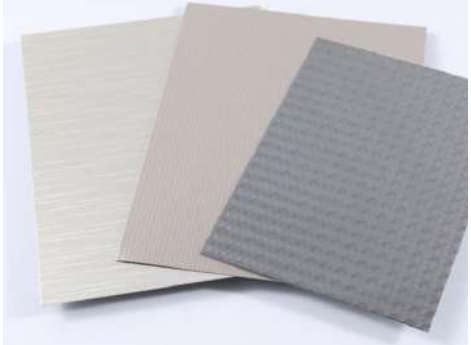
PVC Film Laminated Steel Coils/Sheets for Freezer



AA3004 Aluminium Coated Sheets Used for Metal Roofing Ceiling



Alucosuper Pre-Painted Steel Coil for Home Appliance Panel



VCM-Laminated Steel Coil/ Sheet for Refrigerator

SILICON STEEL SHEET FOR TRANSFORMER

Silicon Steel Sheet for Power Transformer

Stacking Cores *for both Distribution and Power transformers*

Silicon steel Wound Core *for single phase & three phase Distribution transformers*

Amorphous Core *for single phase & three phase Distribution transformers*

Full range of Grain-oriented Electrical Steel thickness from

- ***0.18mm to 0.30mm***
- ***Core loss from 0.65W/kg to 1.4W/kg***

to the Transformer



ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan

Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ENGINEERING TECHNICS

Bussmann by EATON

BUSSMANN - A COMPLETE CIRCUIT PROTECTION SOLUTION















ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan

Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ENGINEERING TECHNICS

Bussmann by EATON

BUSSMANN - A COMPLETE CIRCUIT PROTECTION SOLUTION

Bussmann series IEC cylindrical fuses



Complete portfolio of 8-22 mm diameter Class aM and Class gG cylindrical fuses for industrial applications





Bussmann series Fuses Made Simple™ - Control Circuits

The easiest and fastest way to select and specify the right control circuit fuse



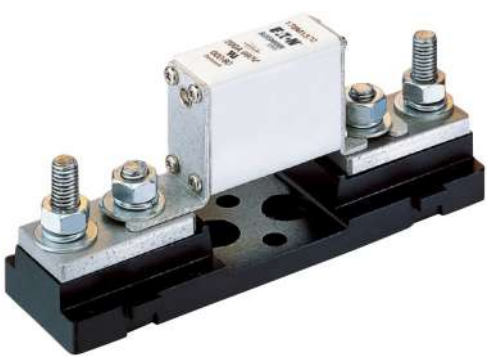
Complete and reliable solar circuit protection













ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan

Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ENGINEERING TECHNICS

NEEQ
Neutral Equipment

NEEQ NEUTRAL GROUNDING RESISTORS (NGR)



Product Range

- *Neutral Point Grounding resistors*
- *Resistors for Wattmetric Current increase*
- *Braking Resistors*
- *Damping Resistors*

- *Voltage range can reach up to 36kV per cascade and power up to some Megawatts.*
- *Advisory services are also offered to customers for optimal use of our products.*

The neutral grounding resistor is rated as follows:

- ***Voltage:*** *Line-to-**neutral** voltage of the system to which it is connected.*
- ***Initial Current:*** *The initial current which will flow through the resistor with rated voltage applied.*
- ***Time:*** *The "on time" for which the resistor can operate without exceeding the allowable temperature rise.*









The product spectrum contains neutral point grounding resistors and resistors for the watt metric current increase, as well as braking and damping resistors.

The voltages can reach approx. 36 kV per cas cade and the power up to some megawatts. We also offer advisory service for the optimal use of our products.

Projects:

2.5Ohm, 90s, 500V, Oil Cooled

2.96Ohm, 3s, 5.77kV Air Cooled

*2*770Ohm, 90s, 11.55kV, Air Cooled*



ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan

Tel: 042-35774702 **Fax:** 042-35774800 **email:** sales@enggtech.net

ENGINEERING TECHNICS

67-A, New Muslim Town, Lahore-Pakistan

Tel: 042-35774702 **Fax:** 042-35774800

email: sales@enggtech.net